

Nexus Regional Dialogues Programme & Energy Efficient Water Sector Project

Development of a Framework for the Governance of the Water, Energy, Food and Environment (WEFE) Nexus in Jordan

Report

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List of abbreviations, local and technical terms

EU	European Union
FAO	(UN) Food and Agricultural Organization
GDP	Gross Domestic Product
GIZ	German Development Cooperation
IEA	International Energy Agency
IRENA	International Renewable Energy Agency
IWRM	Integrated Water Resources Management
JBCO	Jordan Biogas Company
JCC	Jordan Cooperative Corporation
JEPCO	Jordanian Electric Power Company
JPRCO	Jordan Petroleum Refinery Company
JREEEF	Jordan Renewable Energy and Energy Efficiency Fund
JVA	Jordan Valley Authority
MCM	Million Cubic Metre
MED	Mediterranean
MEMR	Ministry of Energy and Mineral Resources
MoENV	Ministry of Environment
MoPIC	Ministry of Planning and International Cooperation
MWI	Ministry of Water and Irrigation
NARC	National Agriculture Research Center
NCP	National Conveyor Project
NEPCO	National Electric Power Company
NERC	National Energy Research Center
NITT	Nexus Intersectoral Technical Teams (Working Groups)
NPC	National Petroleum Company
PHS	Pumped Hydropower Storage
PMD	Programme Management Directorate
RE	Renewable energy
REP	Rural Electrification Project
SG	Secretary General
STC	Steering Committee
TCC	Technical Coordination Committee
WAJ	Water Authority of Jordan
WFP	World Food Programme
WEFE	Water, Energy, Food, Environment (<i>nexus</i>)
YWC	Yarmouk Water Company

Executive summary

To thrive, nations must sustainably manage their resources, including water, food, energy, and the supporting environment (ecosystems and natural resources). Given the interconnectedness of these systems, a water-energy-food-environment (WEFE) nexus approach is crucial for understanding interlinkages and developing synergistic management strategies. This approach recognizes the deep interdependence of resources and the trade-offs that may arise, enabling informed decision-making and the identification of optimal solutions. However, effective implementation of WEFE nexus solutions requires a robust governance framework that considers the local context.

Jordan is among the most water-scarce countries worldwide, with very limited freshwater resources. Its food security is at high risk due to a lack of arable land and rapid urbanization. Additionally, Jordan imports most of its energy as fossil fuels, affecting its energy independence and environmental sustainability. These factors present significant challenges to Jordan's water, food, and energy security, as well as the sustainability of its ecosystems and natural resources. To address these challenges, Jordan must adopt a holistic WEFE nexus approach to integrate its pivotal sectors, minimize trade-offs, and promote effective resource management. A dynamic and comprehensive WEFE governance framework is essential for ensuring the successful adoption and implementation of this approach.

Through co-creation and continuous collaboration with relevant stakeholders from governmental, non-governmental, academia, and private bodies in Jordan and in MENA region, this project has successfully resulted in the establishment of a clear and comprehensive WEFE governance framework tailored to the Jordanian context (see Figure 1).

The approved framework clearly defines the roles and structures of entities that are responsible for governing WEFE-related initiatives, projects, strategies, plans, and visions in Jordan. Further, it follows a clear hierarchical scheme that ensures efficient and evidence-based decision-making on matters relevant to the WEFE nexus. Additionally, the framework prioritizes transparency through the active engagement with various stakeholders from the field of academia, non-governmental organizations, and the private sector.

Recognizing the dynamic and interconnected nature of resources management, the framework is designed to be flexible and adaptable, which is manifested in the involvement of specialized working groups for each process rather than relying on static operational structures. Altogether, the developed framework ought to secure Jordan's sustainable development through promoting the WEFE nexus approach and bridging the gap among relevant key stakeholders and decision-makers in Jordan.

The approved **WEFE Nexus governance framework** includes the introduction of technical and steering committees in addition to other working teams and supporting services as following:

1. A **Steering Committee** dedicated to the Water-Energy-Food-Environment (WEFE) NEXUS is established at the Secretary General's (SGs) level. The committee is chaired by the Secretary General of the Ministry of Planning and International Cooperation (MoPIC) and includes membership from the Secretary Generals of the Ministry of Water and Irrigation (MWI), Ministry of Energy and Mineral Resources (MEMR), Ministry of Agriculture (MoA), and Ministry of Environment (MoE). The committee's primary mission is to define the scope of work and identify key tasks related to the governance of the WEFE NEXUS. Additionally, the committee is tasked with overseeing the fulfilment of governmental commitments directed towards the governance of the WEFE NEXUS including NEXUS institutionalization, executing the obligations outlined in the Economic Modernization Vision Executive Program,

and adopting performance indicators for this purpose. This committee reports to the MoPIC minister who reports to the Prime Minister.

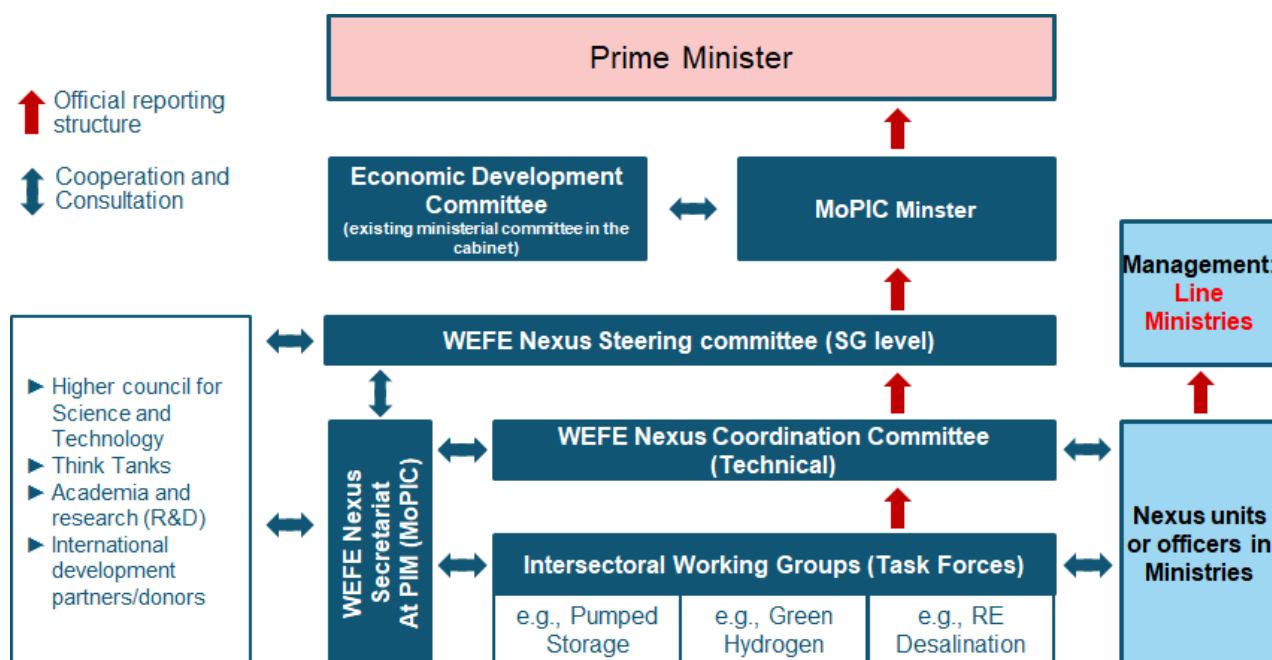


Figure 1. Approved WEFE Nexus governance framework in Jordan

Source: PM office approval

2. A **Technical WEFE Nexus committee** is also established, chaired by the Director of the Public Investment Management Unit, and consists of members from relevant directors/ subject matter experts from MWI, MEMR, MoA, and MoE. The technical committee's mandate encompasses the assessment of the current situation, planning for the WEFE Nexus initiatives, preparation of technical recommendations and their subsequent implementation. Furthermore, the committee is tasked with evaluating various initiatives, highlighting their impact on the relevant sectors and preparing the necessary recommendations. When necessary, the committee may seek the support of experts and consultants to fulfil its tasks.
3. The third entity involved in this governance framework is the **NEXUS Secretariat**, responsible for preparing meeting minutes and proposing fundamental procedures for the WEFE NEXUS. These tasks are assigned to the Supporting Sectors Section/Public Investment Management Unit.
4. **Intersectoral teams** shall be formed according to the nature of the initiative/project following the recommendation of the technical committee submitted to the SGs committee. The primary objective of these sectoral teams is to provide the necessary technical recommendations for the NEXUS institutionalization Process. Additionally, they play a vital role in ensuring the integration of the NEXUS concept into relevant initiatives and projects.
5. The recommendations set forth by the **SGs committee** referenced in item No. 1 are presented to His Excellency the **Minister of MOPIC**. Subsequent to ministerial approval, these recommendations are then submitted to the **Prime Ministry**.

1. Introduction

Water, Food and Energy security are substantial elements of national security in many countries, including the Hashemite Kingdom of Jordan (*hereafter shortly* – Jordan). Ensuring these elements requires sustaining the environment, including ecosystems and natural resources, which are the foundation for these sectors and the overall wellbeing of the country.

In the case of Jordan, the **Water, Energy, Food, and Environment (WEFE) Nexus** poses significant challenges and opportunities. Jordan is one of the most water-scarce countries in the world, with limited freshwater resources and high population growth. The country also relies heavily on imported energy sources and faces energy constraints. Additionally, Jordan's agricultural sector, a major consumer of water and energy, faces various challenges, including limited arable land, climate change impacts, and food security concerns¹. Moreover, the fragile environmental ecosystem in Jordan is vulnerable to degradation due to resource extraction and climate change².

The interdependence and interconnectedness of water, energy, food, and the environment have been recognized as critical components for achieving sustainable development and resource security by many international actors and researchers³. **The Water, Energy, Food, and Environment Nexus approach offers a holistic framework** to understand and address the complex relationships and trade-offs among and within these sectors, where a sound, **integrated governance of the WEFE Nexus** becomes paramount in ensuring the sustainable management and allocation of resources, particularly in water-scarce regions and countries such as Jordan.

Nexus governance refers to the institutions, policies, regulations, and decision-making processes that shape resource management and allocation within the interlinked sectors. Integrated governance is essential for addressing the inherent challenges and complexities and plays a crucial role in managing the WEFE Nexus effectively through coordinated decision-making and collaboration. By adopting an integrated WEFE Nexus governance approach, countries can enhance resource efficiency, minimize conflicts, and maximize the overall benefits for society and the environment.

In Jordan, while the government plays a central role in resource allocation and management, coordination and collaboration among different sectors remain challenging. Barriers to cross-sectoral coordination include limited communication and information sharing between ministries and agencies responsible for water, energy, food, and environmental sectors. Fragmented decision-making and overlapping mandates often lead to inefficiencies and suboptimal outcomes⁴.

It is against this background that the EU and the German government, via the Federal Ministry for Economic Cooperation and Development, provided support to the Government of Jordan in establishing an institutional framework and coordination mechanisms for effective WEFE Nexus governance. The aim of respective work implemented in 2022-2024 was to develop a Framework for the Governance of the Water, Energy, Food and Environment Nexus in Jordan.

This document outlines the WEFE Nexus framework generally supported by key local stakeholders (MOPIC, Prime ministry *et al*). It was elaborated through several rounds of bilateral interviews and

¹ see (WEF Nexus Index, 2022)

² see (AlBatayneh *et al.*, 2022)

³ e.g. Komendantova *et al.*, 2019

⁴ see (GIZ, 2017)

multi-lateral consultations with line ministries, their management and specialists, as well as academia and interested development partners, and discussed in-depth at workshops (the final one was held in Amman on 28 May 2024 – Appendix I) where useful comments were received and taken on board.

The authors believe that results of this work, while addressing the specific challenges and opportunities faced by a water-scarce country like Jordan, would contribute also to the global discourse on the WEFE Nexus governance. And ultimately, they would help evidence-based policy formulation and strategies that promote the sustainable management of the WEFE Nexus, not only in Jordan but also beyond its borders.

2. Project Methodology

The methodology for this project incorporated various approaches at different stages, all involving close coordination with, and participation of, relevant stakeholders (*a participatory, collaborative approach*). It included online and in-person interviews to gather diverse perspectives and insights from representative stakeholders across sectors. Additionally, it included an extensive review of literature to identify global best practices and to assess existing WEFE nexus policies and regulatory frameworks in Jordan. Our approach also included conducting capacity building and engagement sessions to introduce the WEFE nexus approach to cross-sectoral stakeholders and to make available a platform for a dialogue about current challenge and opportunities for developing and operationalizing a WEFE governance framework. An overview of the project methodology and different outputs achieved are summarized in Figure 2 below, followed by a more detailed elaboration.

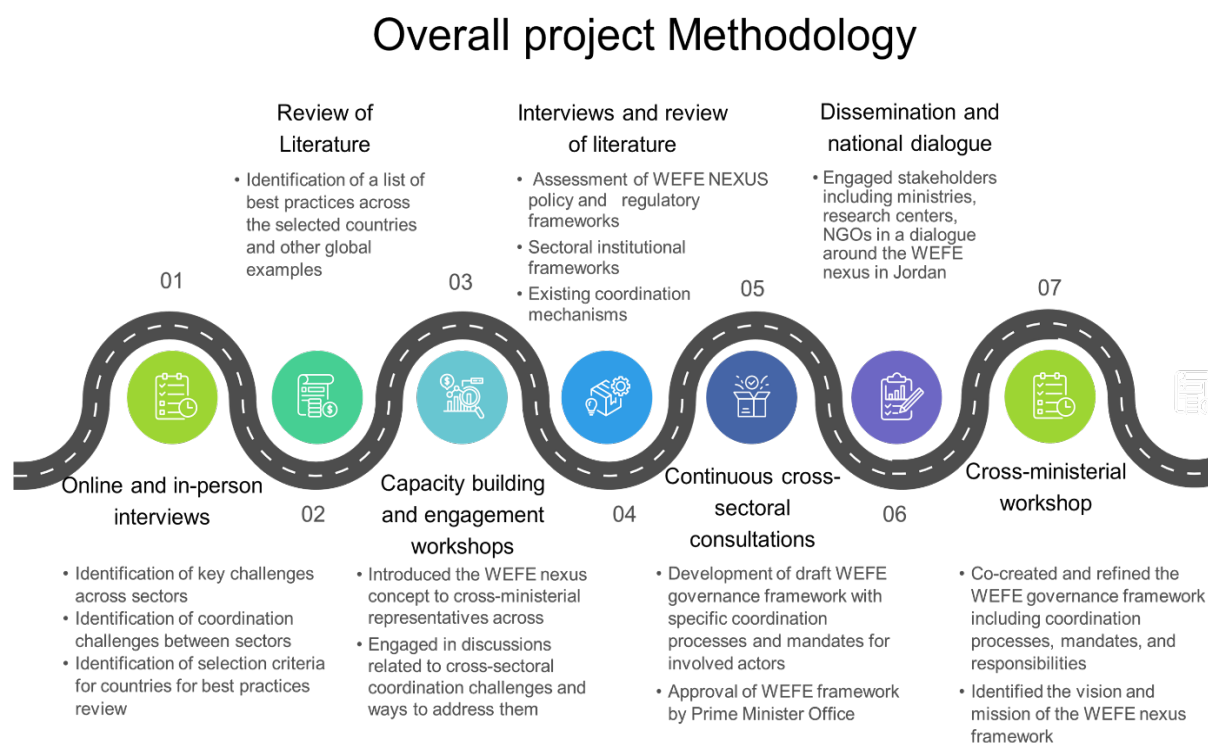


Figure 2. Overall project methodology

Our approach included:

A. Communication and collaboration with stakeholders through:

- i. **Email correspondence.** The GIZ project team developed an initial draft of country selection criteria which was circulated with governmental stakeholders, where feedback and input was requested. The draft included a list of resource availability, climate, economic, financial, and governance criteria, among others. The engaged stakeholders included members of: Ministry of Energy and Mineral Resources, Ministry of Water and Irrigation, Water Authority of Jordan, Jordan Valley Authority, Energy & Minerals Regulatory Commission.
- ii. **Virtual stakeholder consultations.** Following the email request, the consulting team organized a series of virtual meetings with the same stakeholders to learn about the key challenges facing different sectors. These interviews were also used to get further feedback about countries which could offer lessons learned and best practices to be considered in the case of Jordan. These virtual interviews included additional non-governmental and international development stakeholders including EDAMA Association, GIZ-Jordan, German Development Bank, World Bank, EU delegation to Jordan, and Friedrich-Ebert-Stiftung Jordan & Iraq (Figure 3).

Interviewed Stakeholders

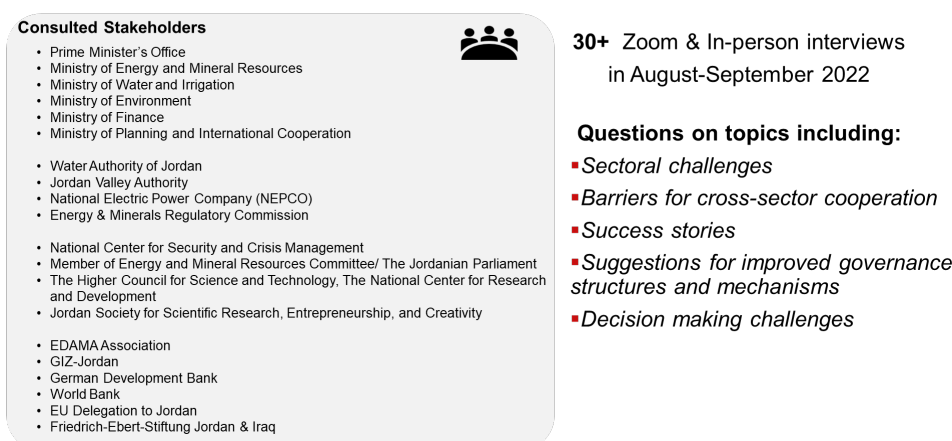


Figure 3. Summary of key stakeholders interviewed through the initial consultation process

B. Studying international good practices and lessons most relevant to the Jordanian context, and transferring them to stakeholders (local capacity development)

- i. **Literature review and analysis.** To identify good practices with WEFE governance and lessons most relevant to the Jordanian context, based on the input provided by stakeholders in response to the country selection criteria and the shortlist of countries to be considered, the consulting team conducted an initial analysis based on reviewing available literature to determine the alignment of the countries of choice with the criteria of interest.
- ii. **Country selection criteria.** Based on the literature review and the feedback from stakeholders, set of criteria was developed to guide country selection. These criteria emphasized similarities in contextual features while also examining a variety of water and energy system characteristics, as well as different economic, governance, and regional cooperation situations. Specifically, it was

determined that the selected countries should be part of, or have similar characteristics to the Mediterranean Sea (MED) basin, and have regions with similarities to Jordan in terms of the challenges they face (see Table 1):

Table 1. Co-created country selection criteria

Water and Energy Resources Criteria (availability, dependence, losses, infrastructure services)	1. Water scarcity 2. Energy dependence 3. Water per capita 4. Energy per capita 5. Non-revenue water 6. Water losses	7. Energy losses 8. Climate (dry, hot with large number of sunny days) 9. Infrastructural development 10. Services coverage (%) 11. Desalination plans 12. Renewable energy plans
Economic/Financial Criteria	1. Level of development 2. Sources of financing 3. Energy costs of the water sector	4. Electricity tariffs differentiated by time of use (TOU) 5. Water cost
Governance Criteria	1. Relevance and how topical for the country is WEN/ existing WEN sector coupling and planning 2. Water and Energy governance structure 3. Role of central government agencies versus basin organizations 4. Operators of energy and water systems (centralized vs. decentralized)	
Regional Cooperation Criteria	1. Neighboring countries' cooperation in water and energy	
Food/Agricultural related Criteria	1. Agriculture water needs 2. Agriculture energy needs 3. The prevalence of desertification, soil erosion, overgrazing	
Environmental Criteria	1. Fresh water quality and quality of treated wastewater	

Source: author's own elaboration

iii. Country selection

During the stakeholder consultation meetings, there was a general agreement that Jordan has several specific peculiarities that would be challenging to find in another country or region. Each country has its unique context in terms of water and energy sources availability, geography, population density and culture, among others. Keeping that in mind, the identified criteria was used to select three countries that have a reasonable level of similarity in terms of challenges, while offering a diverse range of lessons learned and best practices which could be considered for the Jordanian context.

Several countries were initially considered and shared with stakeholders. Following feedback from the consultations, and after collecting information about resource availability, climate conditions, governance structures, and technological availability, among others, the level of similarity or suitability for each criterion was determined as either 1) very similar to Jordan, 2) somewhat similar to Jordan, or 3) far from being like Jordan. The most eligible countries were then discussed until consensus was reached regarding the countries that would be most instructive to examine further. The project team, upon consultation with local stakeholders, determined **Tunisia, Morocco, and Singapore** to be the best candidates for further study based on their similarities and alignment with the criteria. The following sections explore each of the selected countries. Below is a high-level

summary of identified good practices from each of these countries, in addition to other global examples. The table 2 also highlights the key advantages and considerations that need to be accounted for.

Table 2. Summary of good practices

Good Practices	Advantages	Considerations
TUNISIA		
Variable electricity pricing	-Better reflects the true value of energy -Can encourage more efficient patterns of consumption - Reduce peak demand for electricity, which can lead to lower water consumption - Help to encourage the use of renewable energy sources with lower water footprint	-May be unpopular to transition to -Requires efficient metering
Subsidies	-Subsidies can effectively encourage the adoption of new, efficient technologies, which can improve infrastructure and encourage more efficient consumption patterns	-Making one resource more accessible (e.g. energy through solar technology) can lead to easy access and overuse of another resource (e.g. pumped groundwater) unless proper regulations are put in place
Agriculture and water are under one ministry	-This structure can encourage coordination between two interconnected sectors	- Must have mechanisms to ensure actual coordination in practice
MOROCCO		
Inter-ministerial commissions and their role in improved policy coherence	-Councils and commissions coordinate and communicate at the highest level to ensure long-term water strategies are coordinated with other relevant ministries	-These bodies are primarily consultative, political will is required to use their recommendations -There is an absence of coordinating bodies between sectors and areas at the local and technical level
National Office of Electricity and Drinking Water	-The combination of water and energy utilities acknowledges the connection between two sectors and integrates the abundance of institutional knowledge within well-established offices	- Must have mechanisms to ensure actual coordination in practice
Decentralized water governance	- Decentralized water governance allows for more local decision-making and can help to better align water and energy management at the local level. - Facilitate greater stakeholder engagement in decision-making processes to build consensus around water and energy management decisions. - Improve accountability by providing more transparency and opportunities for public participation in decision-making processes.	-Success of decentralized governance requires appropriate national support and resources
Privatization and PPPs	-PPPs and PPAs represent opportunities for efficient concessions and technical expertise	-PPPs and PPAs often include high levels of risk that the government takes on, such as assurances of purchasing all energy produced by private entities. This can be costly to the state.
SINGAPORE		
Clear and actionable coherent strategies with measurable targets	-Clear and measurable targets facilitate intra-governmental and public engagement with transparent measures of success -Coherent and actionable strategies set forth clear priorities for water and energy initiatives to focus on	-Creating measurable and transparent targets requires sound analysis of national capabilities and trajectories

Long-term planning and strict enforcement	-Long-term planning facilitates commitment to infrastructure and plans with long term time frames -Strict enforcement shows commitment and helps achieved intended effects of policies and regulations	-Long-term planning requires sound analysis of national capabilities and trajectories -Strict enforcement of regulations requires political will and monitoring infrastructure -Strict enforcement of regulations may be unpopular
Use of new technology to improve efficiencies	-New technology has the potential to increase efficiencies	- The development of new technologies requires financial and time investments
Good relationship with and ownership by the public	-Public engagement and ownership of water and energy challenges can help with demand management and encourage more efficient consumption patterns	- Requires time and financial resources
United Arab Emirates		
Water and energy under managed and governed under one ministry	-Can catalyze coordination and planning between two interconnected sectors guided by collective outcomes	-Coordination mechanisms need to be in place to ensure meaningful coordination happens
SAUDI ARABIA		
Environment, Water, and Agriculture under one ministry	-Can catalyze coordination and planning between two interconnected sectors guided by collective outcomes	-Coordination mechanisms need to be in place to ensure meaningful coordination happens
SAN ANTONIO, TX, USA		
Addressing legal and procedural barriers through expanding institutional mandates and developing cross-sectoral coordination mechanisms	-More coherent planning and evaluation of interventions through collective indicators	- This will require stakeholder buy-in and investment in financial and human resources
GLOBAL EXAMPLES		
Multi-stakeholder dialogues toward participatory WEFE governance	-Stakeholders from different levels (local, regional, and national) have valuable insights on water and energy challenges -Organized dialogues provide a medium for structured and pointed discussion with clear themes	-Including too many stakeholders without clear processes can diffuse the dialogue and its intended outcomes
Local community participation and bottom-up decision-making mechanisms	-These mechanisms involve the perspectives of the residents who are often most affected by national decisions and often create solutions that are publicly favorable	-Careful organization and planning must go into establishing, supporting, and engaging with local and provincial-level groups
National Policy Dialogues on WEFE issues	- National Policy Dialogues can lead to clear national decisions on complex topics	-The leaders, objectives, and results of the policy dialogue must be clearly thought out and kept in institutional memory - NPD is primarily consultative, political will is required to use its recommendations.

Source: author's own elaboration

vi. Capacity building workshops. During the initial period of in-person visits with stakeholders, we organized a half-day workshop that served as an introduction to the WEFE Nexus approach. Representatives from various ministries were invited to this workshop, where they engaged in a dialogue about the challenges and opportunities for cross-sectoral coordination and the importance of developing a common governance framework. This workshop also provided an opportunity for our team to share insights gathered from the stakeholder interviews, fostering a deeper understanding and collaboration among all participants.

C. Assessing existing WEFE related frameworks in Jordan, identification of opportunities and the development of, and consultations on, the WEFE Nexus governance framework

- i. **Review of literature and stakeholder consultations, to:** 1) map existing WEFE related initiatives and identify WEFE opportunities in Jordan, 2) assess WEFE Nexus policy, regulatory, and institutional frameworks, and 3) identify eventual gaps and conflicts across policies. It included a comprehensive review of the existing reports and other relevant documents and literature on the WEFE Nexus issues and governance in Jordan. It also included interviews and consultations with key stakeholders from government agencies, non-governmental organizations, academia, and the private sector. The assessment focused on identifying strengths and weaknesses in the current governance framework, exploring opportunities for improvement and recommending policy measures to enhance integrated decision-making and resource management. The findings from this assessment contributed to the discourse on the WEFE Nexus governance while addressing the specific challenges and opportunities faced by Jordan.
- ii. **Cross-sectoral consultations that led to the development of the WEFE governance framework:** Based on common reflections gathered through stakeholder consultations and an assessment of WEFE-related institutional, regulatory, and policy frameworks, our team developed a draft of the envisioned WEFE Nexus framework. This framework included key players, a reporting structure, and defined mandates. Through an iterative process of engagement with stakeholders, the framework continuously evolved and was refined. The culmination of these efforts led to the presentation of the finalized framework for approval by the Prime Minister's Office.
- iii. **Continuous cross-sectoral consultations leading to the approval by PM office:** The iterative engagement process involved ongoing cross-sectoral consultations, ensuring that the WEFE Nexus framework addressed the needs and concerns of all stakeholders. Regular feedback loops and collaborative sessions allowed for the incorporation of diverse perspectives and refinements. This continuous dialogue and alignment of interests were crucial in achieving the consensus necessary for the framework's approval by the Prime Minister's Office.
- iv. **Dissemination and national dialogue:** Our team brought together national stakeholders at the 3rd Nexus Conference in March 2024, which served as a platform for sharing progress and innovations across sectors. This conference also facilitated a national dialogue around WEFE governance, highlighting key gaps, challenges, and opportunities in operationalizing such a governance framework. The event fostered a collaborative environment for stakeholders to discuss and align strategic priorities and implementation pathways.

- v. **WEFE ministerial workshop to refine the framework:** To further refine the WEFE Nexus governance framework, our team helped organize and facilitate a ministerial workshop that brought together representatives from relevant ministries. This workshop provided a platform for dialogue to fine-tune the specifics of the WEFE framework and to agree on a common vision and mission moving forward. The collaborative discussions ensured that the framework was robust, actionable, and aligned with the strategic objectives of all participating ministries.

The following sections present the WEFE Nexus governance framework generally supported by ministries (sectoral and MOPIC) and approved by the Prime Minister's Office.

3. Mission and Vision

Vision: The Water-Energy-Food-Environment (WEFE) Nexus is a conceptual framework to achieve the overall welfare of the society and environment through balance of interests and resources. The aim of the WEFE Nexus is to achieve sustainable security of water, energy, food supplies, climate protection, and enhanced societal welfare.

By understanding and managing the interconnections between water, energy, food, and ecosystems, the WEFE Nexus governance system in Jordan strives to achieve sustainable development goals, ensuring resource security and resilience in the face of growing demands and environmental challenges.

Mission:

- Ensure the efficient utilization of resources while protecting the climate
- Mitigate Trade-offs and Build Synergies
- Enhance Resilience of Resource Management
- Promote Informed, Science-based Decision Making
- Ensure Policy Coherence and Integrated Management
- Facilitate Innovative Financing
- Enhance Knowledge Sharing and Capacity Building
- Balance Stakeholder Interests

4. WEF Nexus Governance Framework

4.1 Mandates

Mandates of committees and groups participating in the approved framework

WEFE Nexus Steering Committee

Membership	Mandate	Reporting to
▶ Permanent members: SG of MoPIC, MWI, MEMR, MoA, MoEnv ▶ Chaired by MoPIC SG	▶ Decision making body for WEF Nexus initiatives, trade-offs and synergies. ▶ Define the scope of work and identify key tasks related to the governance of the WEF NEXUS ▶ Endorse and guide proposed Nexus strategies and plans, regulations, solutions, and projects ▶ Lead the task of preparing and implementing institutional and regulatory changes required for the effective governance of the WEF NEXUS ▶ Facilitate securing funding for WEF nexus initiatives	➤ MoPIC Minister

Description:

A committee dedicated to the Water-Energy-Food-Environment (WEFE) Nexus governance is established at the Secretary Generals (SGs) level. The committee is chaired by the Secretary General of the Ministry of Planning and International Cooperation (MoPIC) and includes membership from the Secretary Generals of the Ministry of Water and Irrigation (MWI), Ministry of Energy and Mineral Resources (MEMR), Ministry of Agriculture (MoA), and Ministry of Environment (MoEnv).

The WEF Nexus Steering committee serves as the pivotal decision-making body responsible for overseeing initiatives within the Water-Energy-Food-Environment Nexus domain, managing trade-offs, and leveraging synergies.

The committee's primary mission is to define the scope of work and identify key tasks related to the governance of the WEF Nexus. Additionally, it leads efforts to identify, prepare, and implement institutional and regulatory reforms necessary for effective governance within the WEF Nexus framework.

Furthermore, the committee facilitates the process of securing funding for WEF Nexus initiatives by identifying opportunities, mobilizing resources, and supporting stakeholders in developing funding proposals. Key tasks include reviewing and endorsing proposed Nexus strategies, plans, regulations, solutions, and projects; providing guidance and oversight throughout the lifecycle of Nexus projects; leading the task of preparing and implementing institutional and regulatory changes necessary to enhance governance within the WEF Nexus framework; and actively engaging with donors, financial institutions, and other stakeholders to facilitate the mobilization of funding for WEF Nexus initiatives, ensuring financial sustainability and scalability of projects.

WEFE Nexus Technical Coordination Committee

Membership	Mandate	Reporting to
► Members from MoPIC, MWI, MEMR, MoA, and MoEnv, preferably these members are the nexus officers ► Chaired by the Director of the Public Investment Management Unit at MOPIC	► Report on progress in implementing Nexus initiatives ► Facilitate stakeholder consultations ► Provide recommendations and update steering committee on progress with nexus related projects/ issues ► Support with information and data collection for informed decision-making on WEFE Nexus ► Identify and prioritize WEFE NEXUS initiatives/projects, facilitate their assessment and development ► Coordinating the process of developing integrated strategies.	► WEFE NEXUS Steering Committee

Description:

The SGs committee seeks the assistance of a technical committee, chaired by the Director of the Public Investment Management Unit, and consists of members from relevant directors/ subject matter experts from MWI, MEMR, MoA, and MoEnv.

The WEFE Nexus Steering committee serves as the pivotal decision-making body responsible for overseeing initiatives within the Water-Energy-Food-Ecosystems (WEFE) Nexus domain, managing trade-offs, and leveraging synergies.

The technical committee plays a pivotal role in advancing WEFE Nexus initiatives by undertaking various responsibilities. First, they carefully track and document the progress of ongoing initiatives, compiling comprehensive reports to showcase implementation advancements. Additionally, they foster collaboration and engagement by facilitating stakeholder consultations, ensuring that diverse perspectives are considered. Regular updates and recommendations are provided to the steering committee, keeping them informed about project progress and addressing any emerging issues. Furthermore, the committee supports decision-making processes by gathering and analyzing relevant information and data, enabling informed choices within the WEFE Nexus framework. They also actively identify and prioritize new initiatives, guiding their assessment and development to maximize impact. Ultimately, the committee's coordination efforts ensure the seamless development of integrated strategies, promoting coherence and alignment across all aspects of the Nexus approach.

WEFE Nexus units or officers in Ministries

Membership	Mandate	Reporting to
▶ Officials from MoPIC, MWI, MEMR, MoA, and MoEnv, (preferably same members of the technical coordination committee)	▶ To serve as the First Focal Point in each ministry for other nexus substructures ▶ Identify and Coordinate with all internal structures on nexus related projects/topics within each institution ▶ Provide recommendations and update top management on progress with nexus related projects/ issues ▶ Technical backstopping on nexus related projects/topics	➤ Management within own Ministries

Description:

The Nexus units consist of members from MoPIC, MWI, MEMR, MoA, and MoEnv (preferably same members of the technical coordination committee). The Nexus Officer within each ministry serves as the primary point of contact and coordination for Nexus-related matters, acting as the key liaison between the ministry and other Nexus substructures or coordinating bodies. Their responsibilities entail identifying and engaging with internal structures involved in Nexus-related projects and initiatives, ensuring coherence and collaboration across departments. Regular updates and progress reports provided to senior management enable informed decision-making and strategic guidance on Nexus integration. Additionally, the Nexus Officer offers technical expertise and support to internal teams, facilitating the implementation of best practices and methodologies in the WEFE Nexus domain. Through these efforts, the Nexus Officer plays a crucial role in advancing sustainable and integrated management of water, energy, food, and ecosystems within the ministry's mandate.

WEFE Nexus Intersectoral technical teams (NITT)

Membership	Mandate	Reporting to
▶ shall be formed according to the nature of the initiative/project (examples include Pumped Storage, Green Hydrogen, RE Desalination)	▶ Task-bound and time-bound working group to be established as requirement arise ▶ Assess technical and financial feasibility and evaluate socio-economic benefits of specific Nexus initiatives ▶ Arrange preparing feasibility studies and regulatory requirement studies ▶ Facilitate developing analytics to evaluate the trade-offs associated with different WEFE initiatives and the impact they have on improving national collective outcomes (social, economic, environmental) ▶ Implementing other required ad hoc analytical studies in the WEFE Nexus domain (e.g. on sets of water, food and energy security indicators, on ways to increase resilience of water and energy infrastructure, etc.) ▶ Leading the process of developing integrated strategies and indicators (common KPIs), monitoring and evaluation, and reporting on progress	➤ WEFE NEXUS Technical Coordination Committee

Description:

Intersectoral teams are formed according to the nature of the initiative/project following the recommendation of the technical committee submitted to the SGs committee.

A task-bound and time-bound working group will be established as needed to address emerging requirements efficiently. This group will focus on assessing the technical and financial feasibility of specific Nexus initiatives while also evaluating their socio-economic benefits. Additionally, they will arrange for the preparation of feasibility studies and regulatory requirement assessments to ensure compliance and viability.

Furthermore, the group will facilitate the development of analytics aimed at evaluating trade-offs associated with various WEFE initiatives, considering their impact on improving national collective outcomes across social, economic, and environmental dimensions. They will also be responsible for implementing other necessary ad hoc analytical studies within the WEFE Nexus domain, such as examining sets of water, food, and energy security indicators or exploring methods to enhance the resilience of water and energy infrastructure.

Moreover, this working group will take the lead in the process of developing integrated strategies and indicators, including common Key Performance Indicators (KPIs), to guide monitoring, evaluation, and reporting on progress. By coordinating these efforts, the group will ensure alignment and coherence across initiatives and facilitate informed decision-making within the WEFE Nexus framework.

WEFE NEXUS Secretariat and Support UNIT

Membership	Mandate	Reporting to
▶ Shall be provided by the Supporting Sectors Section at the Public Investment Management	▶ Prepare, call for and organize the Steering Committee and the Technical Coordination Committee meetings ▶ Support the organization of other WEFE NEXUS related meetings as advised by the Technical Committee, and playing a key role in facilitating and moderating effective meetings within the WEFE nexus framework ▶ Assume responsibility for the pre meeting (e.g., agenda, briefing materials) and post meeting (e.g., MoM) documentations ▶ Follow up on actions and outcomes from nexus meetings, ensuring that decisions made during meetings are implemented in a timely manner.	➤ WEFE NEXUS Technical Coordination Committee

Description:

The tasks of the WEFE Nexus Secretariat, including the preparation of meeting minutes, and proposing fundamental procedures for the WEFE Nexus, are assigned to the Supporting Sectors Section / Public Investment Management Unit.

The WEFE Nexus Secretariat is responsible for efficiently managing the organization of both the Steering Committee and the Technical Coordination Committee meetings. This entails preparing and issuing calls for these meetings, as well as coordinating logistical arrangements. Additionally, the Secretariat supports the organization of other WEFE Nexus-related meetings as advised by the Technical Committee, playing a crucial role in facilitating and moderating these sessions within the Nexus framework.

As part of their responsibilities, the Secretariat assumes accountability for both pre-meeting and post-meeting activities. This includes preparing pre-meeting materials such as agendas and briefing documents to ensure all invitees/participants are adequately informed. Following meetings, they meticulously document proceedings, including minutes of meetings (MoM), capturing key decisions, discussions, and action items.

Furthermore, the Secretariat takes charge of following up on actions and outcomes from WEFE Nexus meetings, to ensure that decisions are timely implemented. They collaborate closely with relevant stakeholders to monitor progress, address any obstacles, and facilitate the execution of agreed-upon tasks in a timely manner.

Overall, the WEFE Nexus Secretariat plays a pivotal role in fostering effective communication, coordination, and accountability within the Nexus framework, contributing to the successful implementation of initiatives and the achievement of sustainable development goals.

4.2 Main processes outline

Each operational body within the WEFE governance framework must possess clearly defined roles and responsibilities to achieve an efficient and coherent operational scheme. To clarify this scheme, two main processes are considered in this report which are (i) policy change planning; and (ii) project development.

When planning for a policy change, any stakeholder can initiate a planning process which must be agreed upon by all members of the TCC. Thereafter, the TCC will be responsible for recommending specialized work groups (NITT) along with their mandates to be approved by the STC. Once approved by the STC, the NITT shall conduct a comprehensive review on the sectorial strategies, plans, national visions, and goals to determine the required changes that should be implemented to comply with the WEFE nexus theme. In case a need for change is present, the NITT shall provide the TCC with full documentation outlining the required changes to be reviewed and discussed among the members of the TCC. Once a consensus has been reached among all members on the necessity of amending a policy, the TCC shall forward a detailed report to the STC for approval. Once approved, the NITT will collectively work on implementing the required changes while ensuring a full coherence among national strategies, plans, policies, and national visions. Thereafter, the NITT shall provide the TCC with the updated documents involving the agreed upon amendments. Once the edited documents are agreed upon by all members of the TCC, the role of the NITT can be considered as fulfilled, and the documents must be forwarded to the STC for approval. As a pivotal decision-making entity, the STC shall approve the provided documentation to proceed with the policy planning process. At this stage, the approved documentation by the STC must be directed to the Minister of Planning and International Cooperation, who in turn shall approve amended documents and direct them to the PM for the final approval decision. Once all approvals are granted, the TCC will be responsible for publicly disseminating the amendments, especially for key stakeholders in Jordan. It is worth noting that the consensus of the TCC stands as a pivotal point whenever required, since the presence of a disagreement among the members of the TCC will correspond to the rejection of the sub-outcome provided by the NITT. Further, the entire planning process must be conducted in close coordination with the WEFE Nexus secretariat to ensure coherent operation among all involved entities within the governance framework (Figure 4).

WEFE Nexus Policy Planning

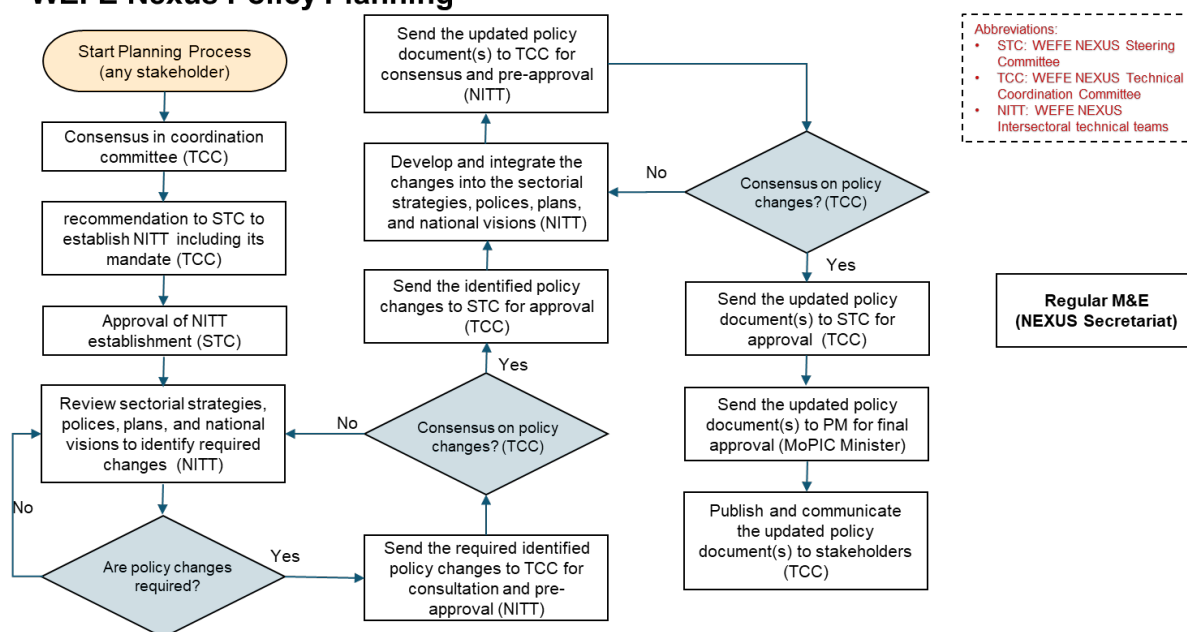


Figure 4: WEFE Nexus Policy Planning

The development of WEFE-related projects/initiatives is another crucial process that can be considered when observing the operational scheme of the WEFE governance framework. In this process, the TCC shall initially identify, preliminary assess, prioritize and shortlist project(s)/initiative(s) to be developed within the WEFE nexus theme. The proposed project(s)/initiative(s) are then sent by the TCC to be approved by the STC. Once the STC approves the necessity for the proposed project/initiative, it shall approve the formation of dedicated NITT working groups based on the recommendations of the TCC. At this stage, the NITT will be responsible for conceptualizing the project/initiative, assessing its feasibility, and finally designing it. Thereafter, the NITT shall register the project/initiative in the National Registry System as a public investment project/initiative, prepare its concept note, Terms of Reference (ToR), and work plan. Further, the NITT should provide the PIM with the project's/initiative's concept note and provide the TCC with the work plan and ToR. If the TCC approves the work plan and ToR, they will be forwarded to the STC for approval. Once approved, the project's/initiative's development shall commence and the NITT will be responsible for supervising and overseeing its progress. Further, the NITT shall prepare regular reports on the progress of the project/initiative and send them to the TCC for review.

The TCC must continuously assess the results demonstrated by the NITT and provide their feedback to ensure an effective and optimal implementation of the project/initiative. Once the project/initiative is finalized, the STC shall approve its outcomes and direct them to the Minister of Planning and International Cooperation. In the case of large-scale projects, the Minister of Planning and International Cooperation shall secure the approval by the PM to actualize the project.

As seen in both processes (Figure 5), each entity within the WEFE governance framework has its unique roles which must be strictly followed to ensure the coherence of the overall operational scheme. Further, the continuous feedback has been given a top priority and must be maintained throughout any process to achieve optimal outcomes while minimizing the gap between relevant bodies and sectors in Jordan. Further, the supervision of the Nexus secretariat ought to accelerate the communication of results and feedback among involved entities, preventing time creep and ensuring the smooth flow of processes when necessary. Altogether, as entities within the framework continue to collaborate and adapt to the governance scheme, the WEFE nexus approach has the potential to secure Jordan's bright and sustainable future.

WEFE Nexus Project Development

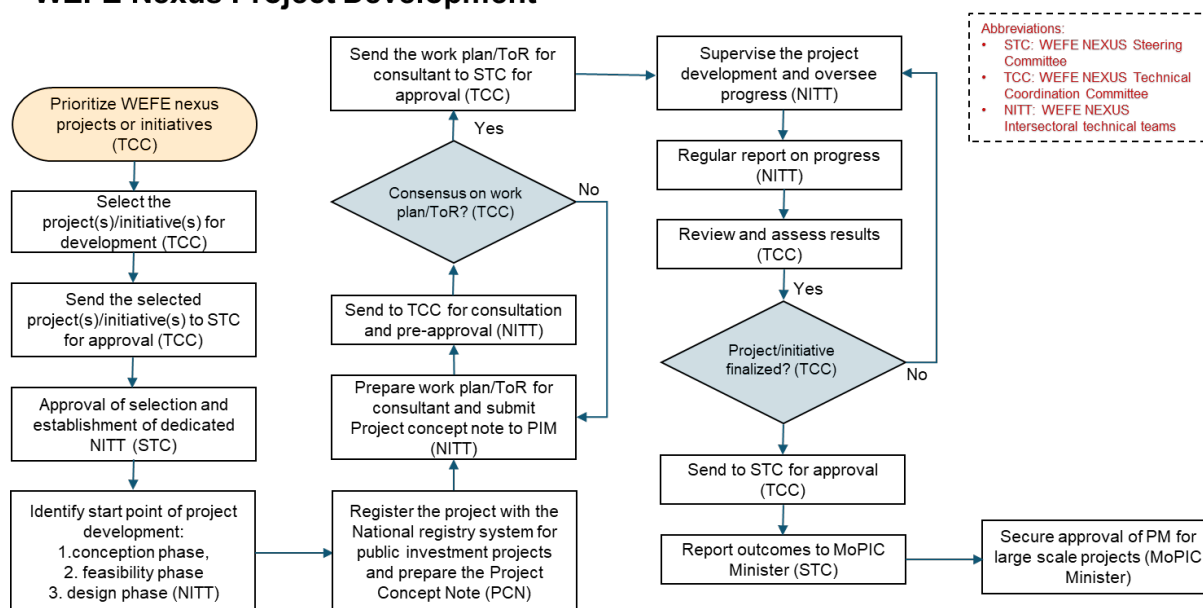


Figure 5: WEFE Nexus Project Development

5. Conclusions on the approved framework

Sustainable development of Jordan significantly relies on effective management of its resources. Especially when considering the challenges facing the country in the water, energy and food security domains, an urgent need for adopting an effective comprehensive WEFE nexus management system becomes evident. It would help address the challenges and explore existing opportunities to ensure the country's sustainable development under the existing constraints: water scarcity, food insecurity, and energy dependency on imports, and the trade-offs among the critical sectors in question.

Recognizing the importance of sound WEFE Nexus governance, and of the first steps already taken by Jordan (such as establishing the inter-ministerial SC at SG level on water-energy nexus), with support of Germany's Federal Ministry for Economic Cooperation and Development, this project was launched to help Jordan create a comprehensive WEFE Nexus governance framework specifically tailored to Jordan's context.

The methodology for this project has been based on close coordination with, and increased participation of, key stakeholders as well as benefiting from global best practices and lessons from the WEFE nexus governance. Another focus was on enhancing the capacity of cross-sectoral stakeholders to ensure a sustainable implementation of the project's outcomes. Initially, the project's consulting team had conducted more than 30 in-person and virtual interviews with relevant stakeholders, mainly domestic and from development partners (DPs) active in the WEFE nexus domain in Jordan. The interviews were conducted with a focus on the opportunities, barriers and challenges related to the adoption of the WEFE Nexus governance framework, while benefiting from the suggestions and recommendations of interviewed officers regarding viable options for governance structures and mechanisms. Thereafter, a comprehensive literature review has been carried out to identify and shortlist countries from around the globe which apply good practices in WEFE Nexus governance. When discussing countries selection criteria, stakeholders stressed on the importance of proximity to Jordan's circumstances with a focus on resources availability, economic conditions, level of international cooperation, governance structure and reliability, water, food and energy sectors conditions, and environmental performance. As a result, it has been determined that Tunisia, Morocco, and Singapore are the most fitting candidates from which success stories and lessons can be learnt.

Local aspects of implementing the WEFE Nexus governance framework in Jordan, and identification of WEFE Nexus-related opportunities, existing initiatives, and relevant frameworks, policies, and regulations along with their gaps was done through: extensive review of available reports and other literature on Jordan, complemented by cross-sectoral consultations with stakeholders in Jordan, including individuals from governmental, non-governmental, academia, and private sector bodies and institutions. These consultations focused on uncovering the potential strengths and weaknesses in the existing governance framework, while providing practical insights on the opportunities for improvement and associated recommendations.

Based on the findings of the review process along with the cross-sectoral consultations, the WEFE governance framework has been developed to address trade-offs and synergies between different WEFE Nexus-related sectors in a holistic manner.

The elaborated WEFE Nexus governance framework has been approved by the Prime Minister office to come into effect. Finally, the feedback of local key stakeholders has been collected and utilized to refine details of the governance framework through two major dissemination events which were the 3rd Nexus conference and the WEFE ministerial workshop.

Altogether, the approved WEFE Nexus governance framework has been developed through profound collaboration and co-creation with key stakeholders from pivotal sectors in Jordan while benefiting from global best practices, success stories and lessons. It follows a hierarchical scheme in which the Prime Minister (PM) sits at the top of the governance pyramid and receives official reports from the Minister of Planning and International Cooperation (MoPIC). Further, the PM will work in close cooperation with the Economic Development Committee which is an existing ministerial committee in the cabinet and will be responsible for the provision of key consultations relevant to the implementation of WEFE initiatives in Jordan. **Over-arching political decisions** on WEFE Nexus will be made at this level.

While the Minister of MoPIC will receive official reports from a WEFE Nexus Steering Committee (STC) which will be chaired by the MoPIC Secretary General (SG) and is composed of the SGs of MoPIC, the Ministry of Water and Irrigation (MWI), the Ministry of Energy and Mineral Resources (MEMR), the Ministry of Agriculture (MoA), and the Ministry of Environment (MoEnv). The STC will be responsible for making **key operational decisions** relevant to the WEFE nexus, identifying synergies and trade-offs, defining the scope of work and securing funding for WEFE nexus initiatives, implementing regulatory changes to promote the WEFE nexus theme, and overseeing policies, strategies, plans, solutions, and regulations, and projects relevant to the WEFE nexus in Jordan.

The STC will receive official reports from a WEFE nexus Technical Coordination Committee (TCC) which will be chaired by the Director of the Public Investment Management (PIM) unit at MoPIC and will consist of members from MoPIC, MWI, MEMR, MoA, and MoEnv. The tasks of the TCC can be summarized in reporting on progress of WEFE nexus initiatives, carrying out consultations with key stakeholders, collecting and refining data relevant to the WEFE nexus theme, identifying and shortlisting potential WEFE nexus initiatives, and coordinating the development of WEFE nexus related strategies, specific projects & initiatives.

Furthermore, the TCC will receive official reports from WEFE Nexus Intersectoral Technical Teams (NITT) which will be formed as required, according to the nature of each WEFE-related initiative or project. These teams will be responsible for assessing the technical and financial feasibility of relevant WEFE projects and initiatives, developing analytics to evaluate the trade-offs and impacts of individual projects and initiatives, and crafting integrated strategies and indicators for the WEFE theme in Jordan.

Both the NITT and TCC will operate in close cooperation with **Nexus Units/Officers** in relevant Ministries, acting as internal focal points for WEFE nexus initiatives and projects. These units/officers will incorporate officials from MoPIC, MWI, MEMR, MoA, and MoEnv and will be responsible for providing consultations and insights to NITT and TCC. Also, each unit/officer will prepare and direct official reports to their respective line Minister.

In addition, a WEFE Nexus Secretariat and Support Unit shall be established at PIM to work in close coordination with, and under the supervision of, the TCC. This unit will be responsible for organizing meetings relevant to the WEFE Nexus among all involved stakeholders, coordinating the exchange of consultations among nexus structures, monitoring and evaluating the progress within the governance framework, providing administrative insights for NITT working groups, and handling public relations and dissemination of information relevant to the WEFE Nexus theme. In parallel, both the STC and the WEFE Support Unit will continuously receive insights from academia and research institutions to ensure the coherence of WEFE nexus-related initiatives and outcomes with the recent scientific advancements in the global WEFE nexus landscape.

Altogether, implementing the approved WEFE Nexus governance framework outlined above is believed to serve Jordanian development efforts towards a sustainable future, greater levels of water, energy and food security while ensuring environmental sustainability and respecting the local

context. Furthermore, it will foster collaboration between water, energy, food, and environment sectors, avoiding siloed decision-making and allowing for a more holistic approach to optimize the management of resources in Jordan.

To build on the established WEFE Nexus governance framework and ensure its effective implementation, several important next steps need to be considered:

First, it is essential to **co-identify common performance indicators** across sectors to facilitate alignment of strategies and enhance coordination. **Developing tools and analytics** to evaluate the impact, synergies, and trade-offs of sectoral plans on these Key Performance Indicators (KPIs) will be crucial for informed decision-making. Additionally, prioritizing **capacity building in systems thinking** will foster a shared understanding of WEFE nexus challenges and opportunities among cross-sectoral stakeholders. Furthermore, implementing **mechanisms for data exchange and sharing** will enable the synthesis of existing data, facilitating varied analyses and quantifying the impact of different strategies on collective indicators.

While the development of a WEFE Nexus framework is an important milestone, ensuring the effective operationalization of such framework will require attention to these steps.

Appendix I: Consultation interviews feedback on approved WEFE Nexus governance framework

Topic	Feedback					
	PM	MoPIC	MWI	MoE	MoA	MoEnv
Comments on approved structure	This is an initial step with possible increased institutionalization in the future. Also, build the capacity of PIM and have them become a proactive Nexus hub in the country.	No Comment	No comments	No Comment	No comments	On the long term it is proposed to establish WEFE Nexus dedicated unit
What Inter-Sectoral Thematic Working Groups should be created, and who are the members?	These should be specialized technical teams from private and public sectors, but the actual themes should be determined by the technical committee.	Working teams to be established as needed	Utilize existing working teams and much before creating new ones	Utilize existing teams and groups as much as possible comprise of public sector employees and private consultants	These should be specialized technical teams from private and public sectors, but the actual themes should be determined by the technical committee.	Based on WEFE nexus concepts to be developed
Is it envisaged that each WEFE ministry establish a Nexus unit internally?	Is not in favor of this, but each ministry can have representatives within the technical committee and the intersectoral thematic working groups.	It is proposed that 2 or 3 focal points be assigned in each ministry and propose to take them on a study tour	Add the role of Nexus to existing units and assign focal points for example (energy unit, water demand, and planning)	Focal points not units	In favor of establishing WEFE Nexus units within each ministry, because Nexus topics are so broad and require several specialties within the same sector.	Focal points in each ministry

What is the frequency of meetings for the SG committee and technical committee?	No specific feedback	As needed	Once a month or as needed	Once each month or as proposed by committee	Proposed 3 meetings per month	As needed
Who will oversee monitoring and evaluation for the WEFE Nexus?	Proposed PIM to become nexus hub and in charge of monitoring and evaluation, but it will require capacity building and strengthening by staffing or hiring consultants.	An outside third party. For example, GIZ team	Develop a list of KPIs and goals and monitor implementation. Also, to liaise with PM public modernization implementation office	An outside third party. For example, GIZ team or private consultant	Proposed PIM to become nexus hub and in charge of monitoring and evaluation, but it will require capacity building and strengthening by staffing or hiring consultants.	Impartial third party
How can transparency and accountability be ensured?	No specific feedback	Develop Minutes of meeting and distribute within committee	Signed minutes of meeting	Develop Minutes of meeting and have them signed and to allow a period of modifications	Issue minutes of meeting which shall be signed by the technical committee members in addition to continuous reporting to the secretary general committee, through a clear operating procedure.	Develop Minutes of meeting and distribute within committee
How to ensure effectiveness and sustainability of the WEFE Nexus governance framework?	Ensure capacity building and strengthening of the PIM unit who are permanent within the MoPIC structure.	Include M&E activities		Monitoring and evaluation in addition to support of senior management	Capacity building of the technical committee members through exposure to successful experiences inside and outside of Jordan.	Issue an official document mandating WEFE Nexus (regulation or policy of strategy,...)

					Develop the mandate of the nexus framework and establish outreach programs within each ministry to ensure everybody knows about it in order to utilize it. Technical committee can invite experts to their meetings as needed.	
How will the WEFE Nexus governance framework be financed?	No specific feedback	MOPEC can try to secure funding	Small part from each ministry and the majority from international donors	Some from each ministry and the majority from international donors	Combined financing with basic function financing coming from the government in addition to support from donors.	Based on themes provide funding suitable for each initiative though MWI and MOPEC
What strategies can be implemented to ensure the long-term success of the framework?	Building the capacity of PIM.	Support the PIM unit by adding a job description related to NEXUS in its mandate	Revise sectorial strategies to ensure they complement each other	Capacity building of the teams though specialized courses and study tours	A study of the existing regulation, strategies and policies should be conducted to identify conflicts. Also, the working members should be compensated for their efforts in addition to their regular salaries.	Sectorial strategies and policies to be reviewed in order to identify and deal with conflicts Issue combined strategies for Nexus initiatives Review all new policies, strategies, and

						regulations with an eye for WEFE Nexus
What mechanisms should be established to resolve potential conflicts between stakeholders or sectors within the WEFE Nexus?	Build the capacity of the PIM to be competent and strong to be able to resolve issues between the stakeholders.	Utilize evidence-based decision making and raise conflicts to higher levels	Conflicts should be documents, properly studies and raised to PM level	Try to consult senior experts from outside the governate sector to give an impartial opinion	Build the capacity of the PIM to be competent and strong to be able to resolve issues between the stakeholders.	Create combined KPIs
What will be the role of Academia and research (R&D) in the proposed structure of WEFE Nexus?	No specific feedback.	Become a thinks tank and can participate in working groups if applicable	Research and capacity building	Conferences, studies, Pilot scale projects and so on	Professionals from academia can participate in technical meetings as necessary. Also, they can participate in capacity building as needed.	Conferences and new technologies

